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GATE ECE · COMMUNICATIONS · CHAPTER-WISE PYQS

Analog Communication Systems

GATE ECE Communications · Previous Year Questions with Answer Key

43	27	1992 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Marks 1	19 questions
Marks 2	24 questions
Answer key	at the end

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Marks 1 19 questions

Q1. GATE ECE 2025

Consider a frequency-modulated (FM) signal

$f(t) = A_c \cos\left(2\pi f_c t + 3\sin\left(2\pi f_1 t\right) + 4\sin\left(6\pi f_1 t\right)\right)$

where A_c and f_c are, respectively, the amplitude and frequency (in Hz) of the carrier waveform. The frequency f_1 is in Hz , and assume that $f_c > 100 f_1$.

The peak frequency deviation of the FM signal in Hz is

(A) $15 f_1$ (B) $12 f_1$ (C) $4 f_1$ (D) $2 f_1$

Open solution

Q2. GATE ECE 2025

Consider a message signal $m(t)$ which is bandlimited to $[-W, W]$, where W is in Hz . Consider the following two modulation schemes for the message signal:

Double sideband-suppressed carrier (DSB-SC):

$f_{\mathrm{DSB}}(t) = A_c m(t) \cos\left(2\pi f_c t\right)$

Amplitude modulation (AM):

$f_{\mathrm{AM}}(t) = A_c(1 + \mu m(t)) \cos\left(2\pi f_c t\right)$

Here, A_c and f_c are the amplitude and frequency (in Hz) of the carrier, respectively. In the case of AM, μ denotes the modulation index.

Consider the following statements:

(i) An envelope detector can be used for demodulation in the DSB-SC scheme if $m(t) > 0$ for all t.

(ii) An envelope detector can be used for demodulation in the AM scheme only if $m(t) > 0$ for all t.

Which of the following options is/are correct?

(A) (i) is TRUE (B) (i) is FALSE (C) (ii) is TRUE (D) (ii) is FALSE

Open solution

Q3. GATE ECE 2024

An amplitude modulator has output (in Volts)

$s(t) = A\cos(400\pi t) + B\cos(360\pi t) + B\cos(440\pi t)$.

The carrier power normalized to 1Ω resistance is 50 Watts. The ratio of the total sideband power to the total power is 1/9. The value of B (in Volts, rounded off to two decimal places) is _____.

Open solution

Q4. GATE ECE 2021

Consider a super heterodyne receiver tuned to 600 kHz . If the local oscillator feeds a 1000 kHz signal to the mixer. The image frequency (in integer) is ____ kHz .

Open solution

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Q5. GATE ECE 2021

Consider a carrier signal which is amplitude modulated by a single - tone sinusoidal message signal with a modulation index of 50%. If the carrier and one of the side bands are suppressed in the modulated signal, the percentage of power saved (rounded off to one decimal place) is ____

Open solution

Q6. GATE ECE 2016 Set 1

A super heterodyne receiver operates in the frequency range of 58 MHz – 68 MHz. The intermediate frequency f_{1F} and local oscillator frequency f_{L0} are chosen such that $f_{1F} \leq f_{L0}$.It is required that the image frequencies fall outside the 58 MHz – 68 MHz band. The minimum required f_{1F} (in MHz) is _____.

Open solution

Q7. GATE ECE 2016 Set 3

For a superheterodyne receiver, the intermediate frequency is 15 MHz and the local oscillator frequency is 3.5 GHz. If the frequency of the received signal is greater than the local oscillator frequency, then the image frequency (in MHz) is _____.

Open solution

Q8. GATE ECE 2015 Set 1

Consider the signal $s(t) = m(t)\cos(2\pi f_c t) + \hat{m}(t)\sin(2\pi f_c t)$ where $\hat{m}(t)$ denotes the Hilbert transform of m(t) and the bandwidth of m(t) is very small compared to f_c. The signal s(t) is a

(A) high-pass signal

(B) low-pass signal

(C) band-pass signal

(D) double sideband suppressed carrier signal

Open solution

Q9. GATE ECE 2014 Set 3

Consider an FM signal $f(t) = \cos[2\pi f_c t + \beta_1 \sin 2\pi f_1 t + \beta_2 \sin 2\pi f_2 t]$. The maximum deviation of the instantaneous frequency from the carrier frequency f_c is

(A) $\beta_1 f_1 + \beta_2 f_2$ (B) $\beta_1 f_2 + \beta_2 f_1$ (C) $\beta_1 + \beta_2$ (D) $f_1 + f_2$

Open solution

Q10. GATE ECE 2010

Consider an angle modulated signal x(t) = $6\cos[2\pi x10^6 t+2\sin(8000\pi t) + 4\cos(8000\pi t)]$ V. The average power of x(t) is.

(A) 10 W (B) 18 W (C) 20 W (D) 28 W

Open solution

Q11. GATE ECE 2008

Consider the amplitude modulated (AM) signal $A_c \cos\omega_c t + 2\cos\omega_m t \cos\omega_c t$. For demodulating the signal using envelope detector, the minimum value of A_c should be

(A) 2 (B) 1 (C) 0.5 (D) 0

Open solution

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Q12. GATE ECE 2005

Which of the following analog modulation scheme requires the minimum transmitted power and minimum channel bandwidth?

(A) VSB (B) DSB-SC (C) SSB (D) AM

Open solution

Q13. GATE ECE 2004

An AM signal is detected using an envelope detector. The carrier frequency and modulating signal frequency are 1 MHz and 2 KHz respectively. An appropriate value for the time constant of the envelope detector is

(A) 500 μsec (B) 20 μsec (C) 0.2 μsec (D) 1 μsec

Open solution

Q14. GATE ECE 2004

An AM signal and a narrow-band FM signal with identical carriers, modulating signals and modulation indices of 0.1 are added together. The resultant signal can be closely approximated by

(A) broadband FM (B) SSB with carrier

(C) DSB-SC (D) SSB without carrier

Open solution

Q15. GATE ECE 2002

A 1 MHz sinusoidal carrier is amplitude modulated by a symmetrical square wave of period 100 μsec. Which of the following frequencies will NOT be present in the modulated signal?

(A) 990 kHz (B) 1010 kHz (C) 1020 kHz (D) 1030 kHz

Open solution

Q16. GATE ECE 1999

A modulated signal is given by, $s(t) = m_1(t)\cos(2\pi f_c t) + m_2(t)\sin(2\pi f_c t)$, where the baseband signal $m_1(t)$ and $m_2(t)$ have bandwidths of 10 kHz and 15 kHz, respectively. The bandwidth of the modulated signal, in kHz, is

(A) 10 kHz (B) 15 kHz (C) 25 kHz (D) 30 kHz

Open solution

Q17. GATE ECE 1998

The image channel selectivity of super heterodyne receiver depends upon

(A) IF amplifiers only (B) RF and IF amplifiers only

(C) Preselector, RF and IF amplifiers (D) Pre selector, and RF amplifiers only

Open solution

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Q18. GATE ECE 1998

A DSB-SC signal is generated using the carrier $\cos(\omega_c t + \theta)$ and modulating singal x(t). The envelope of the DSB-SC signal is

(A) x(t)
(B) $|x(t)|$
(C) only positive portion of x(t)
(D) $x(t)\cos(\theta)$

Open solution

Q19. GATE ECE 1994

$v(t) = 5\left[\cos\left(10^6\pi t\right) - \sin\left(10^3\pi t\right) \times \sin\left(10^6\pi t\right)\right]$ represents

(A) DSB suppressed carrier signal
(B) AM signal
(C) SSB upper sideband signal
(D) Narrow band FM signal

Open solution

Marks 2 24 questions

Q20. GATE ECE 2026

Consider a real, narrowband signal $x(t) = A(t)\cos[2\pi f_c t + \theta(t)]$ where the maximum frequency components of $A(t)$ and $\theta(t)$ are f_M and $f_C\ (= 1000 f_M)$, respectively. Which of the following statements is/are correct for $-\infty

(A) $x(t)$ represents a PSK modulated signal for suitable choices of $A(t)$ and $\theta(t)$.
(B) $x(t)$ represents an amplitude modulated signal for suitable choices of $A(t)$ and $\theta(t)$.
(C) $x(t)$ represents a band-limited Gaussian noise process.
(D) $x(t)$ never represents a narrowband FM signal.

Open solution

Q21. GATE ECE 2023

Let a frequency modulated (FM) signal $x(t) = A\cos(\omega_c t + k_f \int_{-\infty}^{t} m(\lambda)d\lambda)$, where $m(t)$ is a message signal of bandwidth W. It is passed through a non-linear system with output $y(t) = 2x(t) + 5(x(t))^2$. Let B^T denote the FM bandwidth. The minimum value of ω_c required to recover $x(t)$ from $y(t)$ is

(A) $B_T + W$
(B) $\frac{3}{2}B_T$
(C) $2B_T + W$
(D) $\frac{5}{2}B_T$

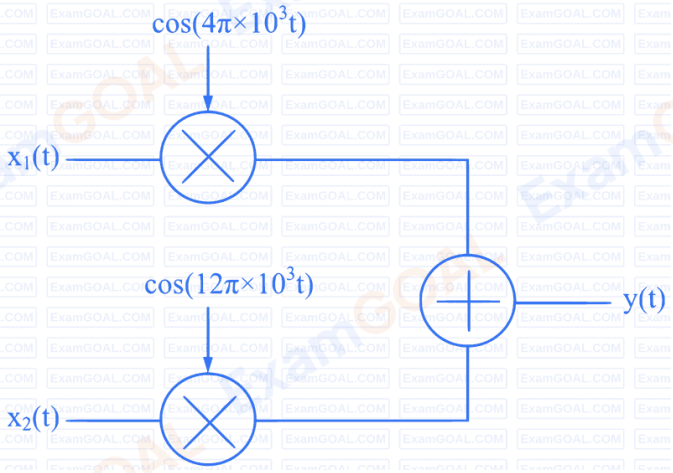
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Q22. GATE ECE 2023

Open solution

Let $x_1(t)$ and $x_2(t)$ be two band-limited signals having bandwidth $B = 4\pi \times 10^3$ rad/s each. In the figure below, the Nyquist sampling frequency, in rad/s, required to sample y(t), is



(A) $20\pi \times 10^3$ (B) $40\pi \times 10^3$ (C) $8\pi \times 10^3$ (D) $32\pi \times 10^3$

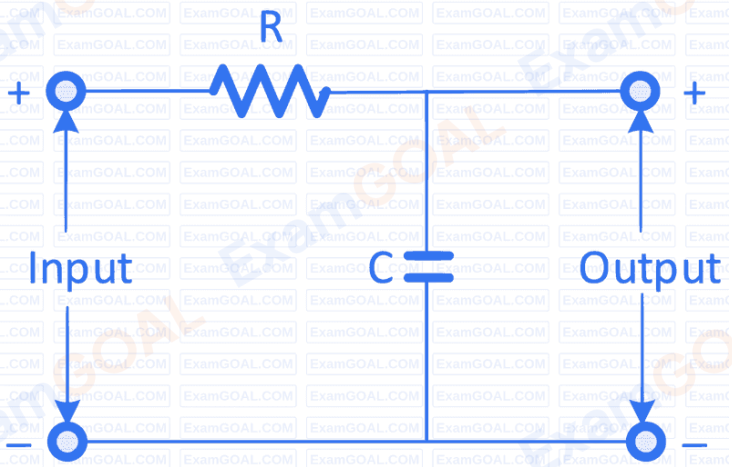
Q23. GATE ECE 2022

Open solution

Consider an FM broadcast that employs the pre-emphasis filter with frequency response $H_{pe}(\omega) = 1 + \frac{j\omega}{\omega_0}$,

where ω_0 = 10^4 rad/sec.

For the network shown in the figure to act as a corresponding de-emphasis filter, the appropriate pairs of (R, C) values is/are _____.



(A) R = 1 kΩ, C = 0.1 μF (B) R = 2 kΩ, C = 1 μF

(C) R = 1 kΩ, C = 2 μF (D) R = 2 kΩ, C = 0.5 μF

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Q24. GATE ECE 2021

Open solution

A sinusoidal message signal having root mean square value of 4 V and frequency of 1 kHz is fed to a phase modulator with phase deviation constant 2rad/ volt. If the carrier signal is $c(t) = 2\cos\left(2\pi 10^6 t\right)$, the maximum instantaneous frequency of the phase modulated signal (rounded off to one decimal place) is ____ Hz.

Q25. GATE ECE 2020

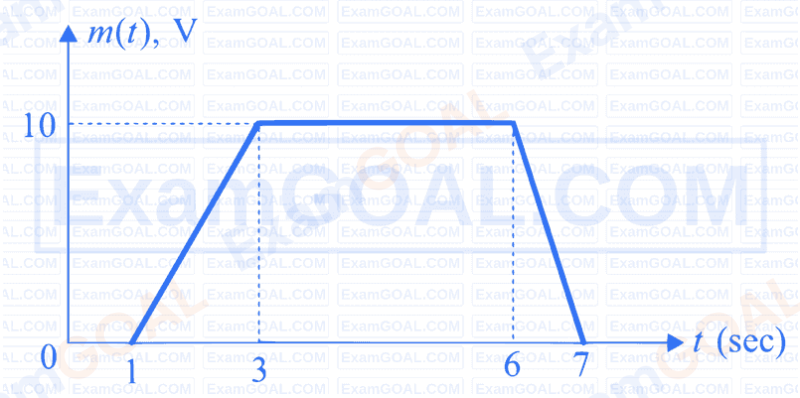
Open solution

$S_{PM}(t)$ and $S_{FM}(t)$ are defined below, are the phase modulated and the frequency modulated waveforms, respectively, corresponding to the message signal $m(t)$ shown in the figure.

$S_{PM}(t) = \cos\left[1000\pi t + k_p m(t)\right]$

$S_{FM}(t) = \cos\left[1000\pi t + k_f \int_{-\infty}^{t} m(\tau)d\tau\right]$

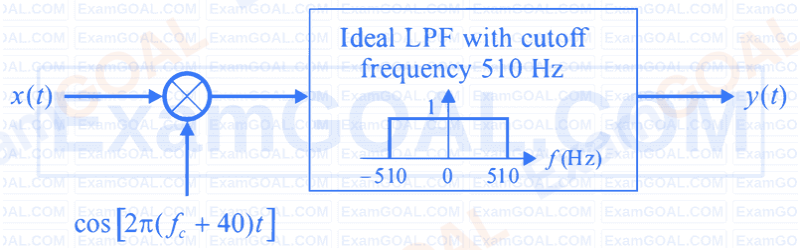
Where k_p is the phase deviation constant in radians/volt and k_f is the frequency deviation constant in radians/second/volt. If the highest instantaneous frequencies of $S_{PM}(t)$ and $S_{FM}(t)$ are same, then the value of the ratio $\frac{k_p}{k_f}$ is ____ seconds.



Q26. GATE ECE 2020

Open solution

For the modulated signal $x(t) = m(t)\cos\left(2\pi f_c t\right)$, the message signal $m(t) = 4\cos(1000\pi t)$ and the carrier frequency f_c is 1 MHz . The signal $x(t)$ is passed through a demodulator, as shown in figure below. The output $y(t)$ of the demodulator is



(A) $\cos(540\pi t)$ (B) $\cos(1000\pi t)$ (C) $\cos(920\pi t)$ (D) $\cos(460\pi t)$

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Q27. GATE ECE 2017 Set 2

The un-modulated carrier power in an AM transmitter is 5kW. This carrier is modulated by a sinusoidal modulating signal. The maximum percentage of modulation is 50%. If it is reduced to 40%, then the maximum un-modulated carrier power (in kW) that can be used without overloading the transmitter is _____

Open solution

Q28. GATE ECE 2017 Set 2

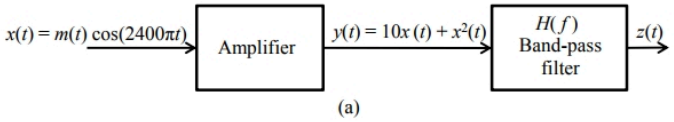
A modulating signal given by $x\left(t\right)=5\sin\left(4\pi10^{3}\mathrm{t}\ -\ 10\pi\cos2\pi10^{3}\mathrm{t}\right)$ V is fed to a phase modulator with phase deviation constant k_p=0.5rad/V. If the carrier frequency is 20 kHz, the instantaneous frequency (in kHz) at t = 0.5 ms is _____

Open solution

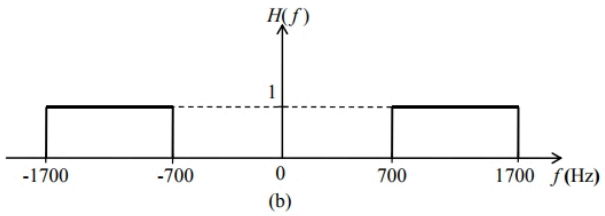
Q29. GATE ECE 2015 Set 1

In the system shown in figure (a), m(t) is a low-pass signal with bandwidth W Hz. The frequency response of the band-pass filter H(f) is shown in figure (b). If it is desired that the output signal z(t) = 10x(t), the maximum value of W (in Hz) should be strictly less than _____.

Open solution



(a)



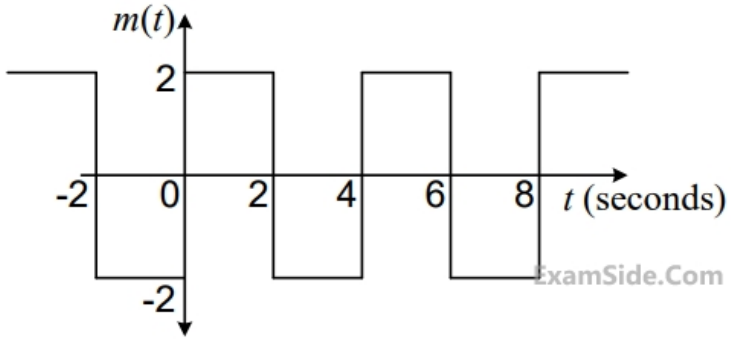
(b)

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Q30. GATE ECE 2012

The signal m(t) as shown is applied both to a phase modulator (with k_p as the phase constant) and a frequency modulator (with k_f as the frequency constant) having the same carrier frequency.



The ratio k_p/k_f (in rad/Hz) for the same maximum phase deviation is

(A) 8π (B) 4π (C) 2π (D) π

Open solution

Q31. GATE ECE 2008

The signal $\cos\ \omega_c t\ +\ 0.5\ \cos\ \omega_m t\ \sin \omega_c t$ is

(A) FM only
(B) AM only
(C) both AM & FM
(D) neither AM nor FM

Open solution

Q32. GATE ECE 2007

A Hilbert transformer is a

(A) non-linear system
(B) non-causal system
(C) time-varying system
(D) low-pass system

Open solution

Q33. GATE ECE 2006

The diagonal clipping in Amplitude Demodulation (using envelope detector) can be avoided if RC time-constant of the envelope detector satisfies the following condition, (here W is message bandwidth and ω_c is carrier frequency both in rad/sec)

(A) $\mathrm{RC} < \frac{1}{\mathrm{W}}$ (B) $\mathrm{RC} > \frac{1}{\mathrm{W}}$
(C) $\mathrm{RC} < \frac{1}{\omega_c}$ (D) $\mathrm{RC} > \frac{1}{\omega_c}$

Open solution

Q34. GATE ECE 2005

A device with input x(t) and output y(t) is characterized by: $y(t) = x^2(t)$. An FM signal with frequency deviation of 90 kHz and modulating signal bandwidth of 5 kHz is applied to this device. The bandwidth of the output signal is

(A) 370 kHz (B) 190 kHz (C) 380 kHz (D) 95 kHz

Open solution

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Q35. GATE ECE 2005

A carrier is phase modulated (PM) with frequency deviation of 10 kHz by a single tone frequency of 1 kHz. If the single tone frequency is increased to 2 kHz, assuming that phase deviation remains unchanged, the bandwidth of the PM signal is

(A) 21 kHz (B) 22 kHz (C) 42 kHz (D) 44 kHz

Open solution

Q36. GATE ECE 2004

Two sinusoidal signals of same amplitude and frequencies 10 kHz and 10.1 kHz are added together. The combined signal is given to an ideal frequency detector. The output of the detector is

(A) 0.1 kHz sinusoid (B) 20.1 kHz sinusoid

(C) a linear function of time (D) a constant

Open solution

Q37. GATE ECE 2003

A DSB-SC signal is to be generated with a carrier frequency f_c = 1MHz using a nonlinear device with the input-output characteristic $v_0 = a_0 v_i + a_1 v_i^3$.where a_0 and a_1 are constants. The output of the nonlinear device can be filtered by an appropriate band-pass filter. Let $v_0 = A_c' \cos\left(2\pi f_c' t\right) + m(t)$ where m(t) is the message signal. Then the value of f_c' (in MHz) is

(A) 1.0 (B) 0.333 (C) 0.5 (D) 3.0

Open solution

Q38. GATE ECE 2003

Let $m(t) = \cos\left[\left(4\pi \times 10^3\right)t\right]$ be the message signal and $c(t) = 5\cos\left[2\pi \times 10^6 t\right]$ be the carrier. c(t) and m(t) are used to generate an AM signal. The modulation index of the generated AM signal is 0.5. Then the quantity $\frac{Total\ sideband\ power}{Carrier\ power}$ is

(A) 1/2 (B) 1/4 (C) 1/3 (D) 1/8

Open solution

Q39. GATE ECE 2003

Let $m(t) = \cos\left[\left(4\pi \times 10^3\right)t\right]$ be the message signal and $c(t) = 5\cos\left[2\pi \times 10^6 t\right]$ be the carrier. c(t) and m(t) are used to generate an FM signal. If the peak frequency deviation of the generated FM signal is three times the transmission bandwidth of the AM singal, then the coefficient of the term $\cos\left[2\pi\left(1008 \times 10^3\right)t\right]$ in the FM signal (in terms of the Bessel coefficients) is

(A) $5\,J_4(3)$ (B) $(5/2)\,J_8(3)$

(C) $(5/2)\,J_8(4)$ (D) $5\,J_4(6)$

Open solution

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Q40. GATE ECE 2000

In an FM system, a carrier of 100 MHz is modulated by a sinusoidal signal of 5 KHz. The bandwidth by Carson's approximation is 1 MHz. If y(t) = (modulated waveform)^3, then by using Carson's approximation, the bandwidth of y(t) around 300 MHz and the and the spacing of spectral components are, respectively.

(A) 3 MHz, 5 KHz (B) 1 MHz, 15 KHz (C) 3 MHz, 15 KHz (D) 1 MHz, 5 KHz

Open solution

Q41. GATE ECE 1993

Which of the following demodulator(s) can be used for demodulating the signal $x(t) = 5\,(1 + 2\ \cos 200\pi t)\cos\ 20000\ \pi t$.

(A) Envelope demodulator (B) square law demodulator

(C) synchronous demodulator (D) None of the above

Open solution

Q42. GATE ECE 1993

A superheterodyne radio receiver with an intermediate frequency of 455 KHz is tuned to a station operating at 1200 KHz. The associated image frequency is _____ KHz.

Open solution

Q43. GATE ECE 1992

The maximum power efficiency of an AM modulator is

(A) 25% (B) 50% (C) 75% (D) 100%

Open solution

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Answer Key

Marks 1

1. **A**	2. **A, D**	3. **2.49 to 2.51**	4. **1400 to 1400**	5. **94.4 to 94.5**
6. **5**	7. **3485**	8. **C**	9. **A**	10. **B**
11. **A**	12. **C**	13. **B**	14. **B**	15. **C**
16. **D**	17. **D**	18. **B**	19. **D**	

Marks 2

20. **A, B, C**	21. **B**	22. **D**	23. **A**	24. **1011313 to 1011314**
25. **2**	26. **C**	27. **5.21**	28. **25**	29. **350**
30. **B**	31. **C**	32. **B**	33. **A**	34. **A**
35. **D**	36. **D**	37. **C**	38. **D**	39. **D**
40. **A**	41. **C**	42. **2110**	43. **B**	

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